

On the Hierarchy Problem of Gravity within the Universal Quantum Foam Hypothesis

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Abstract

The hierarchy problem of gravity is commonly formulated as the extreme discrepancy between the gravitational interaction and other fundamental forces. Within the Universal Quantum Foam Hypothesis (UQSH), this discrepancy is reinterpreted as a scale-dependent manifestation of a single dynamical field rather than as a fundamental fine-tuning problem. Gravity is not treated as an independent interaction with a fixed coupling strength, but as an emergent response of an active space-field whose effective behavior depends on the structural regime of observation. As a consequence, gravitational weakness does not require additional particles, dimensions, or symmetry-breaking mechanisms, but arises naturally from the field's scale-relative organization. This perspective suggests that the hierarchy problem reflects an ontological misinterpretation of gravity rather than a failure of existing physical descriptions.

1 Problem Statement

The so-called hierarchy problem denotes one of the fundamental tensions of modern theoretical physics. It describes the enormous discrepancy between the electroweak energy scale, at which particle masses and interactions of the Standard Model are set, and the gravitational scale characterized by the Planck energy. While electroweak processes take place at energies of approximately 10^2 GeV, the Planck scale lies about 17 orders of magnitude higher. Within quantum field theoretical descriptions, this extreme separation does not appear natural, since corrections due to quantum fluctuations tend to drive mass scales toward the highest available energy scale. The observed stability of the low-energy scales therefore requires, within the standard framework, a finely tuned compensation of large contributions, which is regarded as theoretically unsatisfactory.

In the established interpretation, this problem is understood as an indication that the known theoretical framework is incomplete. Numerous proposed solutions therefore suggest extensions of the particle spectrum, new symmetries, or additional spatial dimensions. What these approaches have in common is that they treat the hierarchy problem as a problem of the relative strengths of fundamental interactions. Gravitation is implicitly regarded as a fundamental force whose exceptional weakness compared to the other interactions requires explanation.

This perspective, however, presupposes that all fundamental interactions are ontologically equivalent and can be directly compared across all scales. In particular, it is assumed that gravitation plays the same physical role on microscopic scales as on cosmological scales, differing only by a very small coupling constant. It is precisely this assumption that lies at the core of the problem considered here. The hierarchy problem arises not only from numerical orders of magnitude, but from the implicit premise that gravitation is a scale-independent, intrinsically defined force that must be treated in the same way at all levels of physical description.

Moreover, the hierarchy problem is closely linked to further open questions, such as the role of gravitation in quantum theory, the significance of the Planck scale as a fundamental limit, and the question of why gravitational effects are practically unobservable in the particle physics regime while they dominate on astrophysical and cosmological scales. These tensions suggest that the problem may not lie solely in missing degrees of freedom or new particles, but in the ontological classification of gravitation itself.

Against this background, the question arises whether the hierarchy problem truly represents a physical fine-tuning problem or whether it emerges from an inadequate conceptual framework. In particular, it must be examined whether the direct comparison between the electroweak scale and the gravitational scale is meaningful if gravitation is not understood as a fundamental interaction, but as an emergent or field-mechanical property of physical space. It is precisely at this point that the Universal Quantum Foam Hypothesis takes effect, by making the assumptions underlying the hierarchy problem itself the subject of analysis.

2 Explanation within the Framework of the Universal Quantum Foam Hypothesis (UQSH)

Within the framework of the UQSH, the hierarchy problem is not interpreted as a numerical mismatch between two fundamental force scales, but as the consequence of a categorical misassignment. The UQSH operates at a deeper level by not treating gravitation as an independent fundamental interaction, but as a field-mechanical property of physical space itself. In this view, gravitation is not a force in the conventional sense, but the response of a dynamic medium to bound, stable, or metastable field states.

Physical space is understood in the UQSH as a continuous quantum foam, an active field medium that exists on all scales in a state of permanent motion, excitation, and relaxation. Matter, radiation, and time are not external actors, but state forms of this medium. Gravitational effects arise wherever field states organize, stabilize, and mutually influence one another. Gravitation is an expression of the internal geometry and tension of the field, not the result of a separate coupling constant.

Within this framework, the notion of a scale-independent gravitational strength loses its meaning. Gravitation possesses no absolute intensity that can be directly compared across all scales. Its effect is always defined relative to the organizational level of the field under consideration and to the structure of the observer. On microscopic scales, where field excitations are high-frequency, shortlived, and weakly coherent, gravitational effects do not appear as a dominant dynamic. In these regimes, the field reorganizes primarily through highly structured couplings such as electromagnetic interactions, which manifest as coherent, directed field excitations with characteristic

selectivity. Gravitation is not absent there, but simply not isolable as an independent effect.

On macroscopic and cosmological scales, by contrast, the same field-mechanical principles act coherently over large distances. Stable material patterns such as stars, galaxies, or galaxy clusters imprint the tension state of the quantum foam over long timescales and large spatial extents. In this regime, gravitation becomes visible as the dominant dynamic, not because it has become stronger, but because the field-mechanical states can organize coherently over sufficiently large scales. The apparent weakness of gravitation in the particle physics regime is therefore not a fundamental puzzle, but a consequence of the scale-dependent readability of field-mechanical effects.

The Planck scale marks the point at which the field can no longer be meaningfully decomposed into independent degrees of freedom, because collective, geometric effects dominate and intrinsic uncertainty, as a constitutive property of stable field states, prevents exact localization. In this context, the Planck scale also acquires a new meaning. It does not mark an energy threshold at which gravitation suddenly becomes strong, but a transition regime in which the description of individual, isolated field excitations loses its validity. The Planck scale marks the transition region in which the temporal and spatial resolution of a given regime reaches its limit and the description of isolated field excitations passes into collective, geometric effects. The hierarchy problem thus arises from the attempt to interpret this boundary as a fundamental force scale rather than as an expression of a regime change in the field description.

From the perspective of the UQSH, gravitation is therefore present on all scales, but not manifest everywhere in the same way. It is not an additional quantity competing with other interactions, but the background mechanism that emerges from the organization of the field itself. The apparent hierarchy of forces does not reflect different fundamental strengths, but different ways in which the field is structured, read, and utilized on different scales.

In this sense, the UQSH resolves the hierarchy problem not by introducing new symmetries, additional particles, or hidden dimensions, but through an ontological reordering. The problem disappears once gravitation is no longer regarded as an isolated force with a fixed coupling constant, but as a scale-dependent property of a dynamic medium that does not recognize a separation between micro and macro. The hierarchy is then not a physical contradiction, but a consequence of the fact that the same field reality appears differently at different levels of organization.

In particular, the results of quantum mechanics and quantum electrodynamics remain unaffected. The model presented here aims to formulate a structural and ontological interpretation of discrete states such as the electron. Where existing theories provide precise predictions, these are not questioned but understood as effective regime descriptions. Within this framework, the UQSH does not claim to replace established theories, but to offer an interpretative extension.

3 Consequences and Outlook

The field-mechanical classification of gravitation within the framework of the UQSH has far-reaching consequences for the interpretation of fundamental problems in physics. If gravitation is no longer understood as an isolated interaction with a fixed coupling constant, but as an emergent property of a dynamic space field, the classical hierarchy problem loses its compelling character. It no longer appears as an indication of an incomplete theory, but as an artifact of a description that forces different scale regimes into the same conceptual categories.

An immediate consequence of this perspective is that the search for a quantized gravitation in the sense of an additional force with an associated exchange particle no longer appears mandatory. In the UQSH framework, quantization does not apply to gravitation itself, but to the field-mechanical excitations and organizational states of the medium from which gravitational effects arise. Gravitation is thereby neither removed nor diminished, but more deeply anchored as a structural property of space, which carries all other physical processes.

Furthermore, this approach suggests that many tensions between particle physics and cosmology arise from the implicit assumption that the same parameters must have the same meaning on all scales. In the UQSH, quantities such as coupling strengths, energy densities, or effective interactions are always to be understood relative to the scale organization of the field. This opens the possibility of interpreting apparent fine-tuning and discrepancies as transition phenomena between different field regimes, rather than treating them as fundamental problems.

For further theoretical development, this implies a clear shift in perspective. Instead of introducing new symmetries, additional particles, or hidden dimensions, the investigation of the field-mechanical properties of space itself moves

to the forefront. Central questions concern the elasticity, relaxation dynamics, stability conditions, and scale-dependent organization of the quantum foam. In particular, the transition regions between micro- and macroscopic regimes gain importance, as they are understood not as singularities or breakpoints, but as continuous reorganizations of the field.

Empirically, this framework also yields testable consequences. If gravitation is a field-mechanical property, gravitational effects should systematically correlate with the stability, coherence, and geometric organization of material structures, rather than exclusively with their mass. This applies to galactic scales as well as to compact objects and early structure formation processes. Deviations from purely mass-based expectations are, in this context, not anomalies, but potential signatures of fieldmechanical organization.

In the long term, the UQSH opens the prospect of a unifying description of physical phenomena without replacing or negating established theories. It provides an ontological framework in which quantum mechanics, relativity, gravitation, dark effects, and cosmic structure formation can be understood as different manifestations of a single dynamic medium. The central challenge lies less in the formal derivation of individual equations than in the consistent reorientation of the underlying concepts.

In this sense, the UQSH is not intended as a conclusion, but as an opening. It invites a reexamination of fundamental problems such as the hierarchy problem not only in computational terms, but ontologically, and encourages taking the properties of space itself seriously as an active component of physical reality.

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